FlowGuard: Smart Camera Traffic Management

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Abdelrahman Ashraf — Mahmoud Yasser — Salah El-deen Mohamed
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Abstract

Traffic congestion serves as one of the most critical challenges due to its consequences for public health, which is rapidly deteriorating over time. It poses serious environmental impacts and challenges: overpopulation, public health, and the pollution of water, soil, and air. The major focus of this project was to tackle the issue of traffic congestion. This project represents a solution focused on developing a smart road system to mitigate heavy traffic, therefore reducing vehicle emissions. In addition, the AI system was used for detecting normal vehicles and emergency ambulance vehicles. The ESP32 development board as a microcontroller and the ESP32-CAM were utilized for object detection of the vehicle in this system. The system's design requirements were the response time, system accuracy, and the trip time. The prototype was designed to handle four critical traffic scenarios. The first case is low traffic that results from a single vehicle on the main road, keeping the green corridor gate closed when the traffic light is red. The second scenario is heavy traffic represented by two vehicles on the main road; then it will send signals to the traffic light to turn green and open the green corridor gate. Moreover, the third case is heavy traffic and the ambulance is coming; the approaching ambulance model overrides the system, ensuring the main traffic light remains red to facilitate emergency vehicle passage. The fourth scenario is low traffic, and an ambulance is coming; the traffic light is still red. Testing the prototype yielded a time response of 1.79 seconds and an accuracy of 98.3%, in addition to a 1.35s trip time. The results successfully met the design requirements, representing the system's potential as a sustainable solution for the urban congestion problem in high-population areas. Modifications were executed across three iterative trials and achieved target values in the final trial.

Introduction

Egypt faces challenges that lead to obstacles on its path toward development and becoming a prosperous nation. Thus, addressing these significant challenges in Egypt, including dealing with urban congestion, reducing pollution fouling our air and water, dealing with population growth, and working to eradicate public health issues **(CH 3.04)**, is the major purpose of this project, which is to tackle the challenge of traffic congestion that relates to these challenges. Traffic congestion has a severe implication for public health and economic stability and numerous serious impacts on the ecosystem **(WHO, 2021)**. In addition, the resultant pollution significantly contributes to the prevalence of diseases that affect human health. Therefore, rapid population growth has precipitated traffic congestion. In the search for good solutions, adaptive control systems were tested. The SCOOT/SCATS system, for example, is a robust and centralized computer-based system for processing the sensor data and tuning the signal timing. However, it is slow in adapting to time updates on fixed time intervals and small increments (such as 4 seconds per minute). In contrast, the SURTRAC system provided a sophisticated decentralized model of the traffic as being a "scheduling problem" and achieved high efficiency (25% reduction in travel time). However, it took vast capital expenditure efforts to equip each intersection with the rather complicated infrastructure (controllers and sensors). This analysis showed the urgent need for a system that addresses the dual challenges of decentralized responsiveness and machine learning capabilities that succeed where previous solutions could not due to their technical and physical limitations. To achieve these important engineering needs, a solution based on an intelligent decentralized control system was chosen and designed. The prototype, which was fabricated on a MDF panel 120 cm x 50 cm, is made out of two integrated control units. The first unit consists of a machine learning based model with 343 vehicle images and ESP32-CAM for real traffic detection and classification. To guarantee an immediate response is highly important, and therefore data is sent wirelessly using ESP-NOW, a fast and weak Wi-Fi type of communication between two ESP32 units, without the risk of depending on a central unique system. This system is particularly designed for the immediate priority needed by emergency vehicles by activating the "green lane" (by moving the servo motor 90 degrees), thus meeting the critical design requirements of saving travelling time and improving traffic efficiency.

Materials

Materials:

Table (1): The purchased materials.

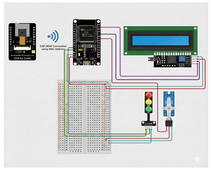
ESP32-CAM CH340 Quantity: (1) Cost: 500 EGP	ESP32 Development Board 38-pin Quantity: (1) Cost: 320 EGP	RC Ambulance Car Quantity: (1) Cost: 360 EGP
Breadboard Quantity: (1) Cost: 35 EGP	LCD I2C Quantity: (3) Cost: 120 EGP	RC Cars Quantity: (2) Cost: 200 EGP
Total Cost: 1915 EGP		
Micro-USB cable Quantity: (2) Cost: 30 EGP	MDF Wood Quantity: (1 sheet) Cost: 150 EGP	Jumper Wires Quantity: (3 packets) Cost: 55 EGP
Small Black Paint Quantity: (1) Cost: 25 EGP	Arduino Traffic Light Quantity: (1) Cost: 35 EGP	Mini Servo Motor Quantity: (1) Cost: 85 EGP

Methods

In order to verify the validity of the proposed project, the prototype was fabricated with an MDF wooden sheet measuring 120 cm in length and 50 cm in width (±2 cm). Two wooden stands dimensioning 12 cm in height each one for the traffic light kit and the other for the LCD I2C. Both were vertically aligned and fixed to the sidewalk using nails for stability; afterwards, the road was painted black with some white dashes in the center of the two lanes, as shown in Figure (1).

Fig(1): The prototype structure.

The traffic management system was executed into two technical phases:
The first phase is a Machine Learning (ML) model which was trained and modified using 343 images of the cars and emergency ambulance car and a Train/Test Split ratio of 86% / 14%, then uploaded as a ZIP library to an ESP32-CAM CH340 to detect, count, and determine the condition of the traffic, as shown in figure (2).
Then, all the readings and outputs were sent by the ESP32-CAM CH340 (sender) to another ESP32 Development Board 38-pin (receiver) **(PH 3.04)** using the MAC Address of the ESP32 Development Board (f4:65:0b:55:a6:e8), as shown in figure (3).



Fig(2): The Diagram shows the ESP, LCD, and traffic light connection.

Test Plan Conditions:

Afterwards, actions by the mini servo motor, traffic light kit, and LCD I2C screen were taken by the instructions of the ESP32 Development Board which they are connected to, for a reason which is that the ESP32-CAM CH340 releases greater power for the operation process of the camera and Machine Learning (ML) model; making the ESP32 Development Board more ideal choice for the project's assembly.

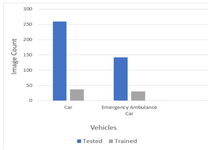
Fig(3): Shows LCD readings and outputs.

The wireless connection between both ESPs is named ESP-NOW which is done over Wi-Fi using a fast, low-power, and wireless communication protocol developed by Espressif **(PH 3.04)**. The commands are restructured based on three conditions. At first, when the ESP32-CAM CH340's Machine Learning (ML) model detects a single car, the traffic light lights red only and the servo motor stays closed while the LCD displays "Cars Detected: 1." Secondly, when the system locates two cars, the traffic light lights yellow for three seconds then green giving them the access to use the green corridor and the servo motor moves 90 degrees upwards with the LCD writing "Cars Detected: 2" then "Go Go Go Access Guaranteed" while detecting the traffic congestion. At last, when an emergency ambulance car is detected, the traffic light lights red with a servo motor standing 90 degrees upwards **(ME 3.02)**, while the LCD shows "Stop" to the cars in the road, then "Emergency, Cars Access Denied" to get the green corridor ready for the high speed of the emergency ambulance car. Moreover, the system with the dependence on the Machine Learning (ML) model prioritizes the emergency ambulance car; as if there are two cars and the emergency ambulance car, the traffic light lights red prohibiting the cars from driving towards the green corridor and levels the servo motor 90 degrees vertically letting the emergency ambulance car drive through the green corridor which saves time for the emergency cases.

Results

After completing the test plan phase, we transferred to collecting and analyzing the results. We carefully analyzed the collected data and summarized it in the following data**:**

As represented in Table (3), the trip time of the road before the green corridor was 2.05 seconds; after constructing and adding the green corridor to the road, the trip time became 1.35 seconds, which is a 34% save of time.
The system has shown an efficiency of 98.3% **(CH 3.01)** with a Train/Test Split ratio of 86% / 14%, as shown in Figure (4) and table (2).



Fig(4): Train & Test split per label.

Table (2): Efficiency according to training images.

	Emergency Ambulance Car	Car
Training	83%	88%
Testing	17%	12%
Total	100% (172)	100% (296)

```
>> PRIORITY: AMBULANCE
Cars: 2 | Amb: 1
>> PRIORITY: AMBULANCE
Cars: 2 | Amb: 1
>> PRIORITY: AMBULANCE
Cars: 2 | Amb: 1
>> PRIORITY: AMBULANCE
```

Fig(5): Ambulance Detected Readings.

Table (3): The Trip Time With the Saved Time Percentage

	Trip Time Without The Green Corredor	Trip Time With The Green Corredor
Time (Seconds)	2.05	1.35
Saved Time (Percentage)	34%	

```
STATUS: CONGESTION DETECTED
Data sent successfully
-----------------------------
STATUS: CONGESTION DETECTED
Data sent successfully
-----------------------------
```

Fig(6): Congestion Detected Readings

Analysis

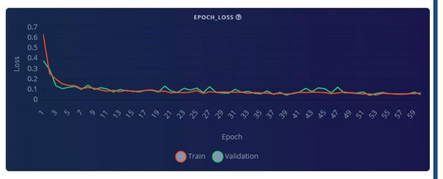
Traffic congestion has been catalogued as a serious factor of air, water and soil deterioration in Egypt. Urban traffic congestion results in increased vehicle idling time and reduced travel speeds, thereby elevating carbon emissions. Additionally, congestion adversely affects air quality and urban development, leading to longer commuting distances, higher congestion costs, traffic accidents, and transportation-related pollution, including climate change, which further exacerbates urban transportation emissions. **(Xiaomei et al., 2025).**

The objectives of the prototype, Internet of Things (IoT), and Information and Communication Technology (ICT) in the adoption to enhance the traffic quality and safety are separated into two tiers:
Tier one: After computing, training, and feeding 343 images to the Machine Learning (ML) model, then uploading the model to the ESP32-CAM CH340; the system was efficiently able to differentiate between normal cars and the emergency ambulance car on the road with the grayscale training settings and efficiency of 98.3% **(CH 3.01)**, as shown in figure (7), which makes the Machine Learning (ML) model capable of detecting traffic congestion, empty roads, and emergency cases.

Fig(7): Objects' detection

Tier two: The detected outputs of the ESP32-CAM CH340 with the Machine Learning (ML) model are sent wirelessly to another ESP32 Development Board 38-pin using ESP-NOW communication protocol **(PH 3.04)** which is fast, smooth, and consumes low-power **(Yaichi et al., 2024);** this ESP32 Development Board is the connection gate between the ESP32-CAM CH340 and the commands of the mini servo motor, LCD I2C, and traffic light kit connected to the ESP32 Development Board after setting the resistance of the LCD I2C potentiometer to get the readings displayed on the screen visibly and clearly, then the mini servo motor's arm is set to angle zero and tested by moving it ninety degrees upwards then returns back to angle zero **(ME 3.02)**.
The Traffic Management System faces both negative and positive effects:
It negatively heats up during the detection process as a result of the high usage of the RAM of the ESP32-CAM CH340 due to the high intensity of the Machine Learning (ML) model's data, as shown in Figure (7), which decelerates the object detection process after a while of running the project's two tiers.
On the other hand, the project has turned into an essential part of the urban passenger transportation system, urban public transportation has played a positive role in alleviating traffic congestion, reducing the demand for fossil fuels and environmental pollution **(MA 3.02)**, and ensuring the accessibility of cities for urban residents (especially urban vulnerable groups including the disabled, the elderly, etc.), thus aiding the sustainable development of cities and urban transport. **(Yuan & Jiaxing, 2022)**
An ambulance car on the road.

Adding the outputs and data collection of the test plan with all that was mentioned earlier:
The Machine Learning (ML) outcome's object detection efficiency rate is 98.3% **(CH 3.01)** on the training and validation set, as shown in Figure (8), after developing, training, and testing 12 previous Machine Learning (ML) models using the ESP32-CAM.



Fig(8): Machine Learning's validation and testing

The application of the Traffic Management System with it's software and hardware has shown a huge enhancement in the traffic flow while dealing with the traffic congestion and its consequences leading to a significant reduction in the fossil fuel consumption **(CH 3.04)** as a result of lowering the trip time; before the usage of the green corridor the trip expenditure was 2.05 seconds and after the usage of the green corridor the trip expenditure shrink to 1.35 seconds; which is a 34% save of time as shown in the following rule, this saved time directly contributed in saving more lives which are driven by the emergency cars from the incident locations to hospitals and clinics.

$$\textit{Trip Time Improvement}: \frac{\textit{Without Green Corridor} - \textit{With Green Corridor}}{\textit{Without Green Corridor}} \times 100$$

Afterwards, the Machine Learning (ML) model's object detection response time was calculated and documented which displayed 1.79 seconds using the normal cars and 2.00 seconds while using the emergency.

While training the Machine Learning (ML) model with it's background and labelling in a certain background, as shown in figure (7).
Therefore, the results in detail show an accuracy of 98.3%, as shown in table (4)

Table (4): Training Rate With Data and Labelling.

	Background	Car	Emergency Ambulance Car
Background	100.0%	0.0%	0.0%
Car	1.9%	98.1%	0.0%
Emergency Ambulance Car	0.0%	0.0%	100%
F1 Score	1.00	0.98	0.99

Table (5): The Learning Outcomes which are utilized in the project.

Learning outcomes	Connection
PH 3.04	Used in designing the communication system between the ESP32-CAM and ESP32 Development Board using ESP-NOW
MA 3.02	Used in measuring and describing the local maxima and local minima of the urban traffic congestion
ME 3.02	Used in understanding the torque of the servo motor's gates
CH 3.01	Used to calculate the accuracy and efficiency of the model's object detection
CH 3.04	Used in understanding the stoichiometry and it's relation with the pollution and public health diseases

Conclusion

Previous solutions provided a valuable starting points, but they were actively sought to address their limitations. Our test plans, analysis, and results showed that our solution is successful in addressing the problem of urban congestion by a smart road system, so it was concluded by comparison between our findings and other prior solutions showed advantages. Solving the urban congestion problem by the SURTRAC project requires hardware at every intersection (computer/processor and detection sensors), which requires a high cost. In addition, the system is only as good as its "eyes." If cameras are blocked by snow, blinded by sun glare, or if sensors fail, the optimization degrades, so it doesn't work effectively anywhere, but our project can be used anywhere at any time because it does not depend on sensors, only ESP and an AI model, so we were able to address the limitations of other prior solutions. Additionally, the goal is achieved by reducing the urban congestion with a time response of 1.79s and an accuracy of 98.3% **(CH 3.01)** in addition 1.35s trip time.

Recommendations

These recommendations aim to boost the efficiency, life time, and performance of the system's applications:
• Smart Community Alert & Help Request System: users can press a button on the mobile application of the smart traffic system for emergencies and urgent help, such as accidents and health issues. Thus, the central server receives the message and identifies the GPS location to recommend the fastest path to reach the person.
• Using a smart weather system: The weather stations connect with the central server of the traffic system to detect heavy trains and extreme weather events. Then, drivers receive the warning message from the mobile app, and AI chooses the best path.
• Car-to-Infrastructure Warning System: The Raspberry Pi can be used in communication with Bluetooth or Wi-Fi because it has a strong communication network and high processing power. Hence, when sudden events and accidents occur, the system automatically sends early warning messages.
• Usage of a Raspberry Pi: The Raspberry Pi offers a faster, smoother, and an easy usage of the Machine Learning (ML) model, which eases the reach of higher efficiency, even the usage of Yolo object detection models easily over the Raspberry Pi's camera.

literature Cited

- Gao, Y., & Zhu, J. (2022). Characteristics, Impacts and Trends of Urban Transportation. Encyclopedia, 2(2), 1168–1182
- Li, X., Wang, G., Zhu, Y., & Liu, W. (2025). A system dynamics-based simulation study on urban traffic congestion mitigation and emission reduction policies. Sustainability, 17(20), 9296.
- Machhi, B., et al. (2023, February). Adaptive traffic control system [Research paper]. International Journal for Research in Applied Science & Engineering Technology (IJRASET), 11(2).
- World Health Organization. (2021, March 9). Road traffic injuries.
- Yaichi, M., Bouchiba, B., & Rebhi, M. (2024). IoT-Based Smart Street Lighting System and Enhanced Energy Saving Using ESP-NOW Protocol. Light & Engineering, 32(6), 76–84.
- Zhou, X., Wang, Y., & Ma, X. (2021). Impact of adaptive lane-management on urban traffic congestion: A micro-simulation study. Journal of Advanced Transportation, 2021, Article ID 6690123.

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For Further Information

abdelrahman.2223029@stemsharkya.moe.edu.eg
mahmoud.2223054@stemsharkya.moe.edu.eg
salaheldeen.2223027@stemsharkya.moe.edu.eg